Graham Felton

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Computer Vision | Generative AI | Applied Machine Learning | Computer Science

Education

Carnegie Mellon University Pittsburgh, PA

M.Sc., GPA: 3.4

- **Awards**: *Academic Merit Scholarship*
- **Relevant Coursework**: *Math for ML, CS for ML, Introduction to Deep Learning, Generative AI, Learning for 3D Vision*

University of Utah Salt Lake City, UT

B.Sc., GPA: 3.65

- **Awards**: *Dean's List (Multiple Years)*

Skills

ML: PyTorch, PyTorch3D, NumPy, OpenCV, WandB, HaMeR, HOLD, Comet.ml
Techniques: Gaussian Splatting (3DGS), Diffusion (DDPM, LoRA fine-tuning), Transformers, 3D Reconstruction, Pose Estimation
Tools: SLURM/HPC, Git, Jupyter, Vim, Pandas, Matplotlib, VS Code
Languages: Python, C++

Experience

Graduate Research Assistant August 2025 – November 2025

Carnegie Mellon University *Pittsburgh, PA*

- Contributed to research towards learning based methods for egocentric pose estimation using prototype device
- Implemented and debugged scripts to optimize visibility of human body given multiview egocentric camera inputs, increasing mean visible area by a factor of 9
- Trained pose estimation model using SLURM GPU cluster for 13 hours/40k iterations achieving PSNR of 28.8 on a novel dataset

Design Technologies Intern May 2025 – August 2025

Payette *Boston, MA*

- Fine-tuned Stable Diffusion via LoRA on custom dataset of designer sketches to replicate firm's hand-sketch aesthetic; model was adopted by visualization staff for sketch-style rendering of conceptual models
- Built Python/PyTorch pipeline for gaussian splat scene reconstruction from Google Street View panoramas of urban contexts; used to produce verified view deliverable for client presentation

Projects

gSLAM | *C++, Eigen* May 2026

- Implemented a probabilistic kinematics and sensing model for a custom robotics simulation from scratch using C++ and Eigen. Implemented EKF SLAM and GraphSLAM algorithms, decreasing robot's current pose estimation error by conditioning pose sample distribution on sensor readings.

Gaussian Splatting (Small Implementation) | *Python, Pytorch, Pytorch3D* October 2025

- (*Coursework for 16-825 Learning for 3D Vision*) Implemented differentiable 3D Gaussian rasterization pipeline in pytorch and trained on randomized scenes with multiple objects

Vision Transformer 2D → 3D Model | *Python, Pytorch, Pytorch3D* May 2025

- Built image to 3D transformer model for architectural model reconstruction from sketch images by fine-tuning DINOv2 encoder, training custom transformer decoder for triplane latent representation; constructed training dataset via automated rendering and sketch-conversion pipeline

Speech Recognition Model | *Python, Pytorch, numpy* March 2025

- (*Coursework for 11-685 Intro to Deep Learning*) Developed Transformer-based ASR system, trained for 148 epochs on SLURM A100 for 38 hours achieving CTC loss of 1.93